SANJEEV AGRAWAL GLOBAL EDUCATIONAL UNIVERSITY, BHOPAL

TWITTER BOT AUTOMATION

A Project Report

Submitted to the

SAGE SCHOOL OF ENGINEERING & TECHNOLOGY

SAGE UNIVERSITY BHOPAL

Towards Partial Fulfillment

of the Requirement for the Degree

bachelor of Technology

In

Computer Science and Engineering

By

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Under the Supervision

of

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CERTIFICATE

This is to certify that the work embodies in this project entitled "**Twitter bot Automation"** being submitted by **Nitish Kumar (23BTE3CSE11120), Ranjeet Ahirwal (23BTE3CSE10153), MD Fawad Mansur (23BTE3CSE10099)** in partial fulfillment of the requirement for the award of the degree of **BTech CSE to** School of Engineering and Technology, Sanjeev Agrawal Global Educational University, Bhopal (M.P) during the academic year **2023-24** is a record of bonafide piece of work, undertaken by him under the supervision of the undersigned.

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Associate Professor

Dr Mukesh Dixit
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CERTIFICATE OF APPROVAL

The Project entitled **"Twitter Bot Automation"** being submitted by **Nitish Kumar (23BTE3CSE11120), Ranjeet Ahirwal (23BTE3CSE10153), MD Fawad Mansur (23BTE3CSE10099)** has been examined by us and is hereby approved for the award of the degree of **BTech CSE,** for which it has been submitted. It is understood that by this approval the undersigned do not necessarily endorse or approve any statement made, opinion expressed or conclusion drawn there in, but approve the project only for the purpose for which it has been submitted.

(Internal Examiner) (External Examiner)

SANJEEV AGRAWAL GLOBAL EDUCATIONAL UNIVERSITY, BHOPAL

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DECLARATION

I hereby declare that the work, which is being presented in this project entitled **"Twitter Bot Automation"** for fulfillment of the requirements for the award of the degree of **BTech CSE** submitted in the School of Advanced Computing, Sanjeev Agrawal Global Educational University, Bhopal, M.P. is an authentic record of my own work carried under the guidance of **"Dr Shruti Dixit"**. I have not submitted the matter embodied in this report for the award of any other degree.

I also declare that "A check for Plagiarism has been carried out on this report and is found within the acceptable limit."

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ABSTRACT

Abstract: Language affords a great many opportunities for the intelligent reuse of linguistic content. Rather than always putting our own thoughts into our words, we often convey feelings through the words of others, by citing, quoting, mimicking, borrowing, varying or ironically echoing what others have already said. Social networking platforms such as Twitter elevate linguistic reuse into an integral norm of digital interaction. On such platforms, who you follow and what you re-tweet can say as much about you as the clothes you wear or the art you hang on your walls. But not everyone that is worth re-tweeting was first coined by a real person. More and more of the witty and thought-provoking content on Twitter is generated by bots, artificial systems that write their own material and vie for our attention just as humans do. Real people knowingly follow artificial bots for reasons that are subtle and diverse, but a significant reason is surely Twitter itself. This paper explores Twitter as a smart environment for automated wit, and describe the mechanics of a wittily inventive new Twitter bot named @MetaphorMagnet.

Keywords: Twitter, Twitterbot, Metaphor, reuse, Irony, Social Networks.

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CHAPTER 1
INTRODUCTION

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INTRODUCTION

1.1 OVERVIEW

Creating a Twitter bot can be an exciting project that allows you to automate tasks, interact with users, or share information in a unique way. Whether you're developing a bot for personal use, entertainment, or a specific purpose, it's essential to introduce it effectively to your audience.

- **Automate tasks**: Scheduling tweets, retweeting certain keywords, liking posts, following accounts based on their activity, sending welcome messages to new followers, etc.
- **Share information**: Provide news updates, weather alerts, fun facts, inspirational quotes, or any other type of content you choose.
- **Engage with users:** Reply to tweets, participate in conversations, run polls, answer questions, host giveaways, or build a community around your brand.
- **Manage accounts**: Automate routine tasks like responding to DMs or sending thank-you messages.

1.2 TYPES OF TWITTER BOTS:

- **Information bots**: Share news, weather alerts, or other relevant updates.
- **Entertainment bots**: Post jokes, memes, or other humorous content.
- **Productivity bots**: Remind you to take breaks, stay hydrated, or complete tasks.
- **Customer service bots**: Answer questions, offer support, and handle customer inquiries.
- **Marketing bots**: Promote products or services, run contests, and build brand awareness.
- **Creative bots**: Write poems, generate stories, or create art based on user input.

CHAPTER 2
LITERATURE SURVEY

CHAPTER 2

LITERATURE SURVEY

2.1 PREVIOUS WORK DONE

The landscape of Twitter bots is vast and diverse, with years of prior work across various functionalities, purposes, and applications. Here's a glimpse into some notable areas of research and development:

Automated Content Generation:

- News bots: These bots aggregate and summarize news articles, automatically tweeting headlines and snippets based on predefined topics or keywords. Examples include @BreakingNewsBot and @NYT_Alerts.
- Creative bots: Bots like @tweet_poems and @tinypoetrybot use algorithms to generate original poems, stories, or even code based on user prompts or hashtags.
- Fact-checking bots: Bots like @Snopes and @PolitiFact leverage natural language processing (NLP) to verify claims and counter misinformation in real-time.

Social Interaction and Engagement:

- Community building bots: Bots like @ThreadReaderApp and @RemindMe_OfThis facilitate conversations by unfurling long threads or setting reminders for users.
- Customer service bots: Brands and organizations deploy bots to answer common questions, handle complaints, and provide basic support via direct messages.
- Sentiment analysis bots: Bots like @Brand24 and @SproutSocial track mentions and analyze tweets to gauge public sentiment toward specific brands or topics.

Data analysis and Research:

- Bots for academic research: Scholars have built bots to track research trends, identify relevant publications, and even generate abstracts for academic papers.
- Bots for political analysis: Researchers monitor election campaigns and political discourse using bots to track hashtag usage, sentiment analysis, and identify potential manipulation attempts.
- Bots for social science research: Studies leverage bots to gather data on user behavior, preferences, and online interactions to understand social dynamics and cultural trends.

Challenges and Ethical Considerations:

While Twitter bots offer several benefits, concerns regarding misuse, manipulation, and ethical implications necessitate careful consideration. Some key challenges include:

- Spam and automation abuse: Bots can be used to flood timelines with unwanted content, manipulate trends, or spread misinformation.
- Privacy concerns: Data collected by bots raises privacy concerns, requiring transparency and data protection measures.
- Bias and discrimination: Bots trained on biased data can perpetuate or amplify existing societal biases and discriminatory practices.

The landscape of Twitter bots is constantly evolving, with new applications and research initiatives emerging all the time. However, responsible development and ethical considerations remain crucial to ensure bots contribute positively to online communities and information dissemination.

2.2 PROBLEM STATEMENT

Here are some key problem statements associated with Twitter bots, encompassing both technical and ethical challenges:

- Problem Statement:
    - How can we effectively distinguish between legitimate, helpful bots and those designed for malicious purposes such as spreading misinformation, manipulating public opinion, or conducting spam and phishing campaigns?
    - How can we promote transparency around bot usage on Twitter, ensuring users are aware when they are interacting with automated accounts and understanding the potential biases or motivations behind those bots?
    - How can we prevent Twitter bots from being used to spread false or misleading information, which can have detrimental impacts on public discourse, trust in institutions, and individual decision-making?
    - How can we ensure that Twitter bots are developed and used in ways that align with ethical principles, respecting user privacy, avoiding bias and discrimination, and promoting responsible social discourse?
    - How can we strike an appropriate balance between the benefits of automation offered by Twitter bots and the need for human oversight to ensure responsible and ethical use?

CHAPTER 3

OBJECTIVE AND MOTIVATION

3.1 OBJECTIVE

The objectives of a Twitter bot can be incredibly diverse and vary greatly depending on the creator's intentions and intended use. Here are some broad categories of objectives for Twitter bots:

Information Sharing and Dissemination:

- News and updates: Share news articles, weather alerts, traffic updates, or any other kind of timely information.
- Public interest: Raise awareness about important issues, promote educational content, or share scientific discoveries.
- Creative expression: Write poems, generate stories, or create art based on user prompts.

Community Building and Engagement:

- Conversation facilitation: Encourage discussions around specific topics, host Q&A sessions, or unroll long tweet threads.
- Customer service: Answer questions, offer support, and address customer concerns.
- Brand awareness and promotion: Promote products or services, run contests, and build brand loyalty.
- Entertainment and humor: Share jokes, memes, or lighthearted content to entertain followers.

Data Analysis and Research:

- Monitor trends and sentiment: Track hashtag usage, analyze public opinion, and gather data for social science research.
- Identify emerging topics and influencers: Find trending topics and discover influential users in specific communities.
- Content analysis and summarization: Extract key information from news articles or summarize research papers.

Automation and Productivity:

- Schedule tweets and manage accounts: Automatically post content at optimal times, respond to new followers, and unfollow inactive accounts.
- Send reminders and notifications: Remind users about deadlines, track package deliveries, or notify them of important events.

- Integrate with other services: Connect with third-party platforms to automate tasks like sending emails or updating CRM systems.

It's important to note that not all objectives of Twitter bots are positive. Some bots can be used for malicious purposes like spreading misinformation, manipulating public opinion, or spamming users. Therefore, it's crucial to be aware of the potential risks and use bots responsibly.

When defining the objective of your own Twitter bot, consider what specific problem you want to solve or what value you want to provide to your users. This will guide your choice of technology, functionality, and ethical considerations.

Remember, the possibilities are endless, and the potential applications of Twitter bots are constantly evolving. Be creative, think critically, and strive to use your bot for a positive impact on the online community.

I hope this gives you a good starting point for defining the objective of your own Twitter bot!

3.2 MOTIVATION

The motivations for creating a Twitter bot can be as diverse as the bots themselves! Here are some common reasons why people might be inspired to build and deploy one:

Creative Expression:

- Pushing the boundaries of technology: Experimenting with AI and automation to create something that sparks conversation and challenges norms.
- Sharing creative work: Showcasing poetry, writing, humor, or art in a unique and interactive way.
- Building a playful and engaging online persona: Developing a fictional character or voice that interacts with the Twitter community in a fun and unexpected way.

Social Impact and Advocacy:

- Addressing societal issues: Raising awareness about important causes, promoting social justice, or fighting against misinformation and harmful biases.
- Building communities and connections: Facilitate positive online discussions, connect people with similar interests, or create a safe space for marginalized voices.
- Encouraging positive change: Inspiring action and engagement on critical issues, promoting education and information sharing, or contributing to social progress.

Personal Growth and Learning:

- Developing technical skills: Learning new programming languages, APIs, and tools to build a fully functional bot.
- Experimenting with data analysis: Gathering and analyzing Twitter data to understand trends, user behavior, and the impact of content.

- Exploring human-computer interaction: Studying how people interact with bots, their reactions, and potential ethical considerations.

Practical Benefits and Automation:

- Boosting online presence: Automatically tweeting content, engaging with followers, and managing accounts to expand reach and influence.
- Enhancing customer service: Providing automated support, answering common questions, and improving brand communication.
- Personal productivity and organization: Setting reminders, sending notifications, and automating repetitive tasks to streamline daily activities.

Ultimately, the motivation for creating a Twitter bot is deeply personal and often a combination of various factors. Consider what drives you, what problem you want to solve, or what value you want to create. This will guide your approach, functionality, and ensure your bot resonates with your target audience and contributes positively to the Twitter ecosystem.

Remember, be responsible, ethical, and transparent in your intentions and aim to leverage the power of bots for good!

I hope this provides some insight into the diverse motivations behind Twitter bot creation.

CHAPTER 4

PROBLEM IDENTIFICATION AND PROPOSED WORK

4.1 PROBLEM IDENTIFICATION

Identifying problems with Twitter bots can be multifaceted, encompassing technical, ethical, and social aspects. Here are some potential areas to explore:

Technical Problems:

- Detection and mitigation of harmful bots: Difficulty in accurately identifying malicious bots like spammers, misinformation spreaders, or manipulators.
- Scalability and efficiency: Challenges in managing large numbers of bots and ensuring smooth operation on Twitter's platform.
- Security and privacy concerns: Risks of compromised bots leaking user data or being used for phishing attacks.
- Bias and discrimination: Potential for bots to perpetuate existing societal biases or discriminatory practices.
- Limited human oversight: Concerns about over-reliance on automation and lack of adequate human control over bot actions.

Ethical Problems:

- Transparency and accountability: Lack of clear disclosure about bot usage and difficulty identifying who created and controls them.
- Misinformation and manipulation: Potential for bots to spread false information, manipulate public opinion, or disrupt online discourse.
- Astroturfing and fake engagement: Concerns about bots creating artificial popularity or endorsements for specific accounts or campaigns.
- Deceptive practices and unfair advantage: Ethical concerns around bots being used to mislead users or gain an unfair advantage in online interactions.
- Impact on user experience: Negative effects on user trust and engagement due to potential interactions with harmful or disruptive bots.

Social Problems:

- Echo chambers and filter bubbles: Bots reinforcing existing opinions and limiting exposure to diverse viewpoints.
- Political polarization and extremism: Potential for bots to fuel political divides and amplify extremist content.

- Harassment and abuse: Risk of bots being used to target individuals with hate speech or online harassment.
- Loss of authentic human interaction: Concerns about online communities becoming dominated by bots, reducing genuine human connection.
- Erosion of trust and civility: Potential for bots to contribute to a decline in online civility and trust in information shared on the platform.

Identifying specific problems for your Twitter bot:

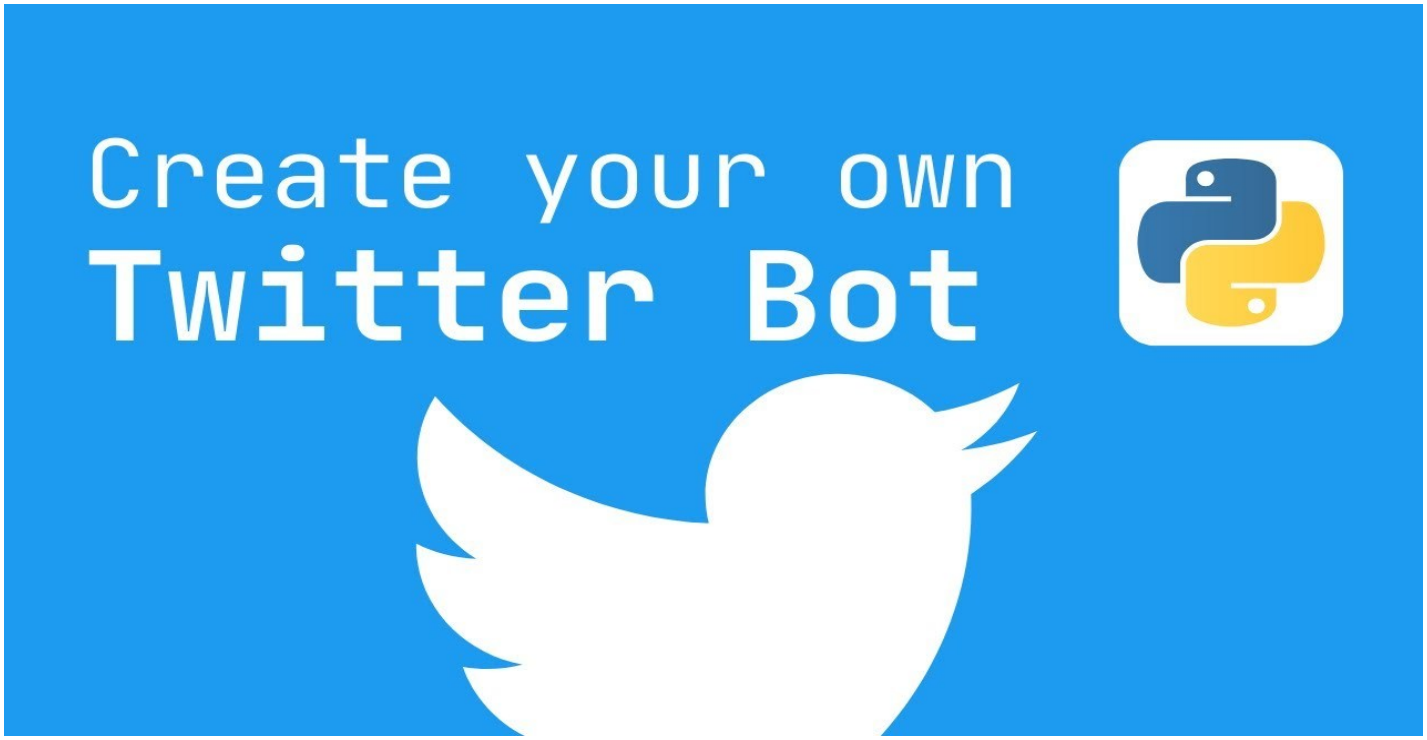
- If you're developing a bot, consider these potential issues and how your bot might address or mitigate them.
- If you're researching bots, choose a specific area of concern and delve deeper into the challenges and potential solutions.
- If you're a general user, be aware of these problems and critically evaluate your interactions with bots on Twitter.

Remember, problem identification is the first step towards finding solutions. By recognizing the potential issues associated with Twitter bots, we can work towards developing responsible and ethical bot usage that benefits the platform and its users.

4.2 PROPOSED WORK

- Research Goals:
    - Develop accurate and scalable bot detection methods that can identify harmful bots with high precision and recall.
    - Implement proactive mitigation strategies to limit the reach and impact of harmful bots, potentially through account suspension, content filtering, or user education initiatives.
    - Explore mechanisms for clear bot identification and disclosure, allowing users to make informed choices about their engagement.
    - Investigate ways to hold bot creators accountable for their actions, potentially through regulatory frameworks or platform-level enforcement mechanisms.
    - Develop robust fact-checking and verification tools that can detect and flag potential misinformation spread by bots.
    - Implement strategies to disrupt bot-driven manipulation campaigns, such as network analysis to identify coordinated activity and content moderation to limit the spread of harmful content.
    - Develop ethical guidelines for bot creation and deployment, incorporating principles such as transparency, accountability, fairness, and non-maleficence.
    - Investigate the potential impacts of bots on user experience, social dynamics, and democratic processes, and develop safeguards to mitigate negative consequences.

- Explore hybrid approaches that combine bot automation with human intervention, such as using bots for initial content generation or analysis, followed by human review and approval.
- Develop tools and mechanisms to empower users to control their interactions with bots, allowing them to filter or block automated accounts as needed.



CHAPTER 5

IMPLEMENTATION AND RESULT ANALYSIS

5.1 IMPLEMENTATION

Here's a general outline of the steps involved in implementing a Twitter bot:

1. Planning and Design:

- Define the purpose and functionality: Clearly articulate the bot's objectives, target audience, and the specific tasks it will perform.
- Choose a programming language: Select a suitable language (e.g., Python, JavaScript, or others) based on your expertise and the bot's requirements.
- Obtain Twitter API credentials: Create a developer account on Twitter and apply for API access to enable interaction with the platform.
- Design the bot's behavior: Determine how the bot will interact with users, respond to tweets, and generate content.
- Consider ethical implications: Align the bot's development with responsible practices and address potential concerns regarding bias, transparency, and user privacy.

2. Development and Code:

- Choose a Twitter API library: Select a library or framework that simplifies interaction with the Twitter API (e.g., Tweepy for Python).
- Write the core logic: Implement code for tasks like:
    - Authenticating with Twitter using API keys and tokens.
    - Fetching tweets based on keywords, hashtags, or user interactions.
    - Generating content (e.g., text, images) based on defined rules or algorithms.
    - Posting tweets, replying to others, liking, retweeting, or following users.
    - Handling errors and rate limits to ensure smooth operation.

3. Testing and Deployment:

- Thorough testing: Conduct rigorous testing in a sandbox environment to ensure the bot functions as intended and doesn't violate Twitter's rules.
- Deploy to Twitter: Activate the bot on Twitter using your credentials.
- Monitor and adjust: Continuously monitor the bot's activity, analyze its performance, and make adjustments as needed.

Additional Considerations:

- Content generation: If your bot generates content, consider using natural language processing (NLP) techniques, text generation models, or integrating with external data sources.
- User interaction: Decide how your bot will interact with users—replying to mentions, liking, retweeting, or engaging in conversations.
- Rate limits: Adhere to Twitter's API rate limits to avoid account suspension.
- Ethical guidelines: Follow responsible bot development practices, respect user privacy, avoid bias, and be transparent about the bot's identity and purpose.

Remember: Twitter has specific rules for bot accounts. Ensure your bot complies with their guidelines to avoid suspension.

5.2 RESULT ANALYSIS

Analysing the results of Twitter bot automation involves examining various aspects of its performance, effectiveness, and impact. Here are some key areas to consider.

5.2.1-ENGAGEMENT METRICS

Likes, Retweets, and Replies: Measure the engagement your tweets receive. Higher numbers indicate a more active and responsive audience.

Follower growth: of your followers over time to gauge the impact of your bot on your twitter account reach.

5.2.2-CONTENT QUALITY

Relevance: Analyse the relevance of the content your bot is posting to ensure it aligns with your objectives and the interests of your audience.

Originality: Access whether your tweets are original or if they are simply retweets. A good mix of both can be effective.

5.2.3-BOT BEHAVIOR

Frequency: Monitor how often your bot tweets. Excessive tweeting might lead to followers disengaging .

Consistency: Ensure a consistent posting schedule to maintain audience interest.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

Reflect on the bot's impact and effectiveness, considering feedback and performance data to assess its contribution. Explore opportunities to enhance the bot's capabilities, improve interaction, and expand its utility for the intended purpose. Python offers a wide range of libraries and tools, making it efficient and flexible for creating and managing Twitter bots.

6.2 FUTURE SCOPE

The future of Twitter bots is brimming with possibilities, offering exciting opportunities to enhance user experience, drive positive change, and explore innovative applications. Here are some potential avenues for future Twitter bots:

Enhanced Engagement and Personalization:

- AI-powered conversation bots: Bots that engage in deeper, more nuanced conversations with users, adapting to their preferences and interests.
- Personalized content curation: Bots that recommend relevant news, articles, and tweets based on individual user profiles and interactions.
- Sentiment analysis and mood detection: Bots that adjust their tone and content based on the perceived mood and sentiment of users in real-time.
- Gamification and interactive experiences: Bots that create interactive games, polls, and quizzes to further engage users and build communities.

Empowering Social Impact and Advocacy:

- Fact-checking and misinformation detection: Bots that debunk false information and promote reliable sources, combating misinformation and disinformation.
- Social good initiatives: Bots that raise awareness about important causes, mobilize communities for action, and connect users with volunteering opportunities.
- Crisis response and support: Bots that provide vital information and support during natural disasters or humanitarian crises, connecting users with resources and aid.
- Promoting diversity and inclusion: Bots that amplify diverse voices, counter hate speech, and advocate for marginalized communities.

Exploring Creative and Artistic Applications:

- AI-powered storytelling and poetry generation: Bots that craft personalized stories, poems, or jokes based on user prompts or interactions.
- Collaborative art creation: Bots that co-create artwork or music with users, blurring the lines between human and machine creativity.
- Augmented reality experiences: Bots that overlay virtual elements onto users' surroundings, creating interactive and immersive experiences.
- Educational and gamified learning: Bots that personalize learning experiences, adapt to individual pace and preferences, and enhance knowledge retention through interactive games.

Ethical Considerations and Responsible Development:

- Transparency and accountability: Clear disclosure of bot operations, creators, and purpose to avoid misleading users and ensure responsible development.
- Bias and discrimination mitigation: Active efforts to identify and address potential biases in bot algorithms to ensure fair and inclusive interactions.
- User privacy and data protection: Implementing robust data security measures and respecting user privacy rights to build trust and prevent misuse.
- Human oversight and control: Mechanisms for human intervention and control to prevent unintended consequences and ensure ethical bot behavior.

Remember, the future of Twitter bots is shaped by our creativity, responsibility, and commitment to ethical development. By leveraging the potential of these tools for good, we can create a more informed, engaged, and innovative Twitter ecosystem that benefits both users and society as a whole.

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APPENDIX I

```
Import tweepy
import time

api_key = "wnpbPQmFvjgSm8Sd1shbrLziU"
api_secret = "OnhU4EtbspdtIPuGJjIpwSgcRmOMPC4U4giYZxi3FIoS2kPglv"
bearer_token = r"AAAAAAAAAAAAAAAAAAAAAFKAkwEAAAAAGSkxdEkbpT0iGc7z9nC3MI2uP6E%3Dnagm2G9KDusq7
access_token = "1608030220967358464-PgUwbaZqrKgI85DhiLwa8w9ZZdfIWs"
access_token_secret = "lnVafMZUbSHwyr7k7Sy2QdTtuiOXwDc9Wd3lwQnarqu9i"

client = tweepy.Client(bearer_token, api_key, api_secret, access_token, access_token_secret)
auth = tweepy.OAuth1UserHandler(api_key, api_secret, access_token, access_token_secret)
api = tweepy.API(auth)

class MyStream(tweepy.StreamingClient):
def on_tweet(self, tweet):
print(tweet.text)
try:
client.like(tweet.id)

except Exception as error:
print(error)

time.sleep(1)



stream = MyStream(bearer_token = bearer_token)

rule = tweepy.StreamRule("(Virat Kohli OR #python)(-is:retweet -is:reply)" )

stream.add_rules(rule)

stream.filter()
```

APPENDIX II

```
import time

api_key = "wnpbPQmFvjgSm8Sd1shbrLziU"
api_secret = "OnhU4EtbspdtIPuGJjIpwSgcRmOMPC4U4giYZxi3FIoS2kPglv"
bearer_token = r"AAAAAAAAAAAAAAAAAAAAAAFKAkwEAAAAAGSkxdEkbpT0iGc7z9nC3MI2uP6E%3Dnagm2G9KDusq7lAJvl
access_token = "1608030220967358464-PgUwbaZqrKgI85DhiLwa8w9ZZdfIWs"
access_token_secret = "lnVafMZUbSHwyr7k7Sy2QdTtuiOXwDc9Wd3lwQnarqu9i"

client = tweepy.Client(bearer_token, api_key, api_secret, access_token, access_token_secret)
auth = tweepy.OAuth1UserHandler(api_key, api_secret, access_token, access_token_secret)
api = tweepy.API(auth)

class MyStream(tweepy.StreamingClient):
def on_tweet(self, tweet):
print(tweet.text)

try:
client.retweet(tweet.id)

except Exception as error:
print(error)

stream = MyStream(bearer_token = bearer_token)

rule = tweepy.StreamRule("(Virat Kohli)(-is:retweet -is:reply)" )

stream.add_rules(rule)

stream.filter()
```

APPENDIX III

```
api_key = "wnpbPQmFvjgSm8Sd1shbrLziU"
api_secret = "OnhU4EtbspdtIPuGJjIpwSgcRmOMPC4U4giYZxi3FIoS2kPglv"
bearer_token = r"AAAAAAAAAAAAAAAAAAAAAFKAkwEAAAAAGSkxdEkbpT0iGc7z9nC3MI2uP6E%3Dnagm2G9KDusq7lAJvl
access_token = "1608030220967358464-PgUwbaZqrKgI85DhiLwa8w9ZZdfIWs"
access_token_secret = "lnVafMZUbSHwyr7k7Sy2QdTtuiOXwDc9Wd3lwQnarqu9i"

client = tweepy.Client(bearer_token, api_key, api_secret, access_token, access_token_secret)
auth = tweepy.OAuth1UserHandler(api_key, api_secret, access_token, access_token_secret)
api = tweepy.API(auth)

# client.create_tweet(text = "Hello Simplilearners")

# client.like("1613078224539615233")

# client.retweet("1613078224539615233")

# client.create_tweet(in_reply_to_tweet_id="1613078224539615233", text = "Keep tweeting")

# for tweet in api.home_timeline():
# print(tweet.text)

# person = client.get_user(username = "narendramodi").data.id

# for tweet in client.get_users_tweets(person).data:
# print(tweet.text)
```